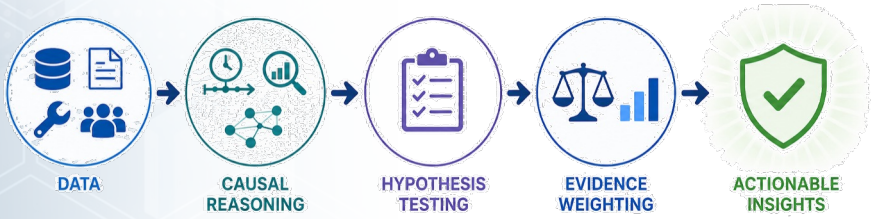
INL/MIS-26-92853



Why Did It Fail?
An Evidence-Based Approach to RCA

Plant Optimization Update and Planning Meeting - June 9th, 2026

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Battelle Energy Alliance manages INL for the
U.S. Department of Energy's Office of Nuclear Energy

Idaho National Laboratory

Context: Root Cause Analysis (RCA)

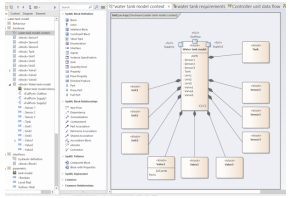
- Manual RCA is hard: Time pressure, system complexity, and heterogeneous data (logs, SOEs, CRs, WOs, telemetry)
- It fails in predictable ways
  - Investigative fixation: converging early on the wrong cause
  - Incomplete coverage: missing contributing and root causes
  - Inconsistent use of history: repeating known failures

Unstructured data



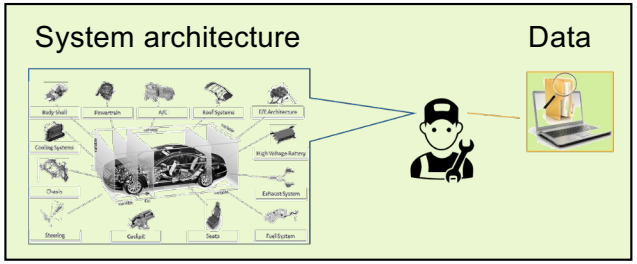
condition reports, work orders, engineering evaluations, procedures, past RCAs

Structured data



system diagrams, condition-based monitoring data, FMEA tables, logs, SOEs

- **This project:** Co-pilot engineers to improve RCA for complex events by combining plant data, and system knowledge into a unified reasoning workflow
- **RCA is first a knowledge management exercise**
  - Retrieve large amounts of data
  - Annotate relations between data elements
  - Extract knowledge from data elements
- **Model-Based Systems Engineering (MBSE) diagrams**
  - Provide context to data: Need to emulate system engineers' mindset
  - Provide the semantic role of each data element

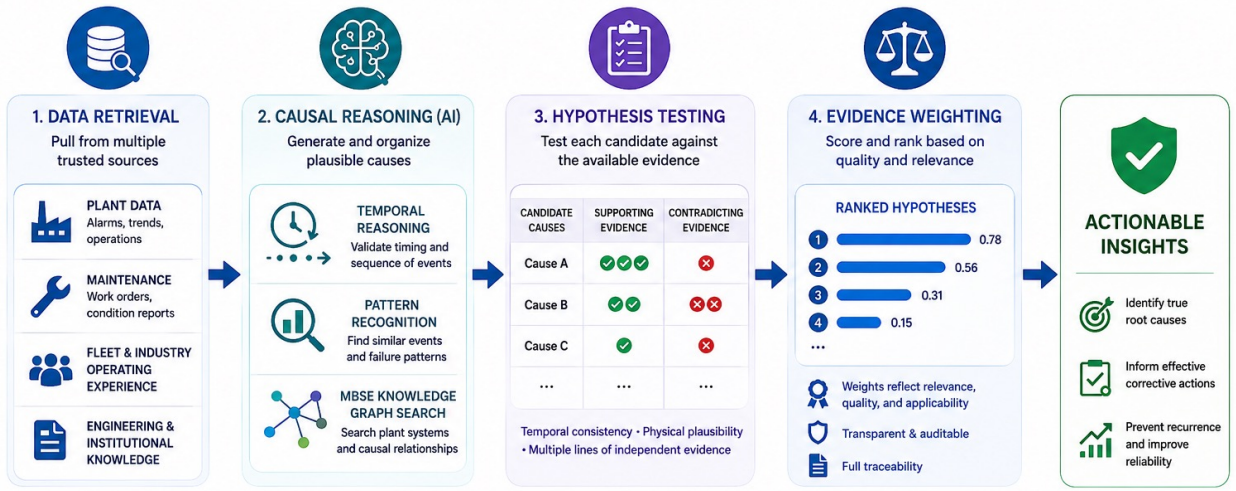


FY26 Objectives and Milestones

- **Goal:** Develop the foundation of a plant **knowledge management and discovery platform** for evidence-based decisions
- **Delivered:** 10-stage pipeline implemented along with 3 structured validation cases completed
  - Knowledge augmented generation (KAG) architecture that integrates data and system architecture
    - Knowledge graph: Digital historian by integrating system architecture (MBSE models), and plant data
    - Chroma store: Document chunks from plant's retrieved records (hybrid retrieval: similarity with keyword matching)
    - Set of data analytics engines
- **Mindset:** Gather data → Hypothesize → Test → Decide

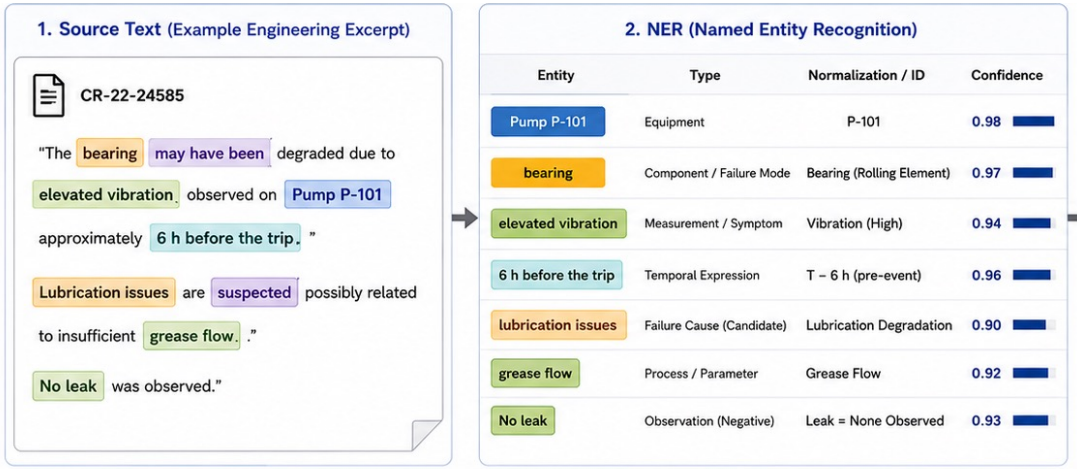
Mirror how system engineer's mindset
- Identify what is possible
- Check what is temporally consistent
- Check what is supported by evidence

1. **Structural reasoning:** Constrain the causal search space based on system architecture
2. **Temporal reasoning:** Identify temporal relationships and patterns among data elements
3. **Evidence-based reasoning:** Retrieve relevant historical records and test against hypothesis



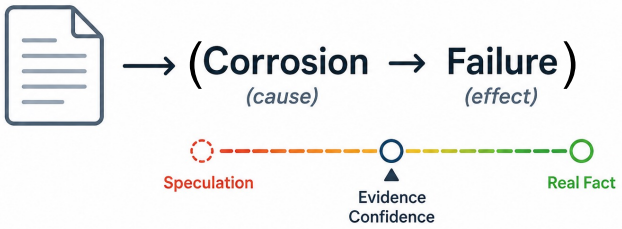
Analytics Methods I: Knowledge Extraction (What Happened?)

- **Named entity recognition:** Pipeline designed to extract entities from retrieved text
  - Equipment and alarm IDs, failure modes, degradation phenomena, temporal and location references, physical measurements



- **Conjecture detection:** Focus on detection of Modal verb density (e.g., may, could, possibly, might)

- **Text causality extraction:** Identify causal relationships from text

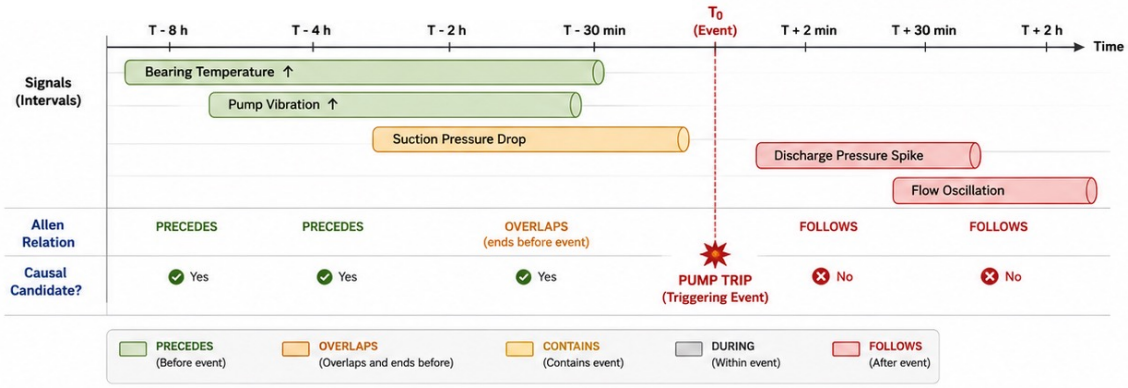


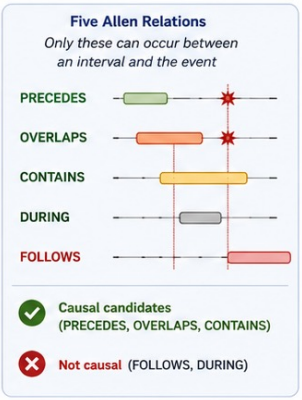
- **Evidence epistemics:** Separate facts from speculation

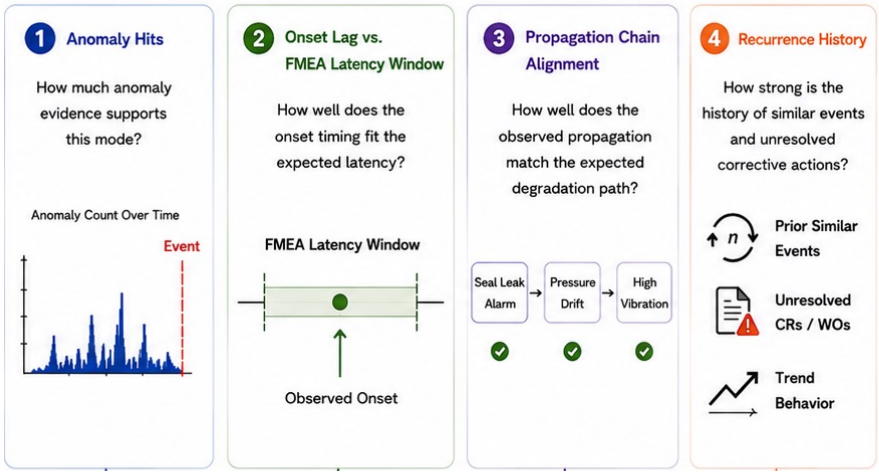
- **PM compliance:** Assess maintenance effectiveness

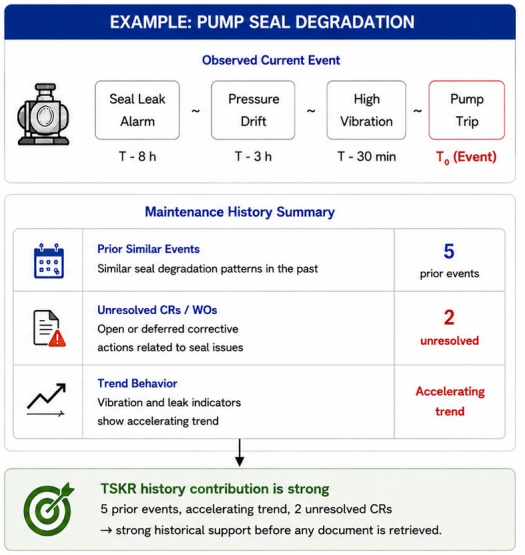
Analytics Methods II: Temporal Data Analytics

- **Temporal analysis** (cause or consequence?): Determine whether events occurred in a temporally consistent sequence leading to the failure
- **Evaluate temporal consistency** (could this phenomena have caused that event?): Compare signal timing and maintenance history against expected failure behavior



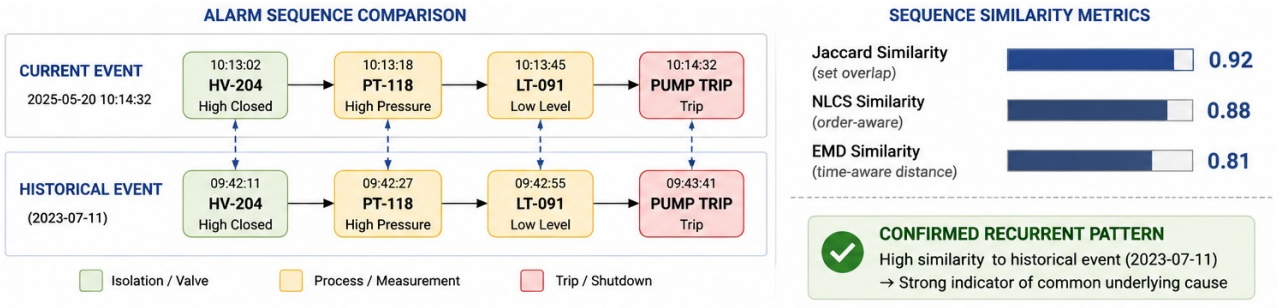




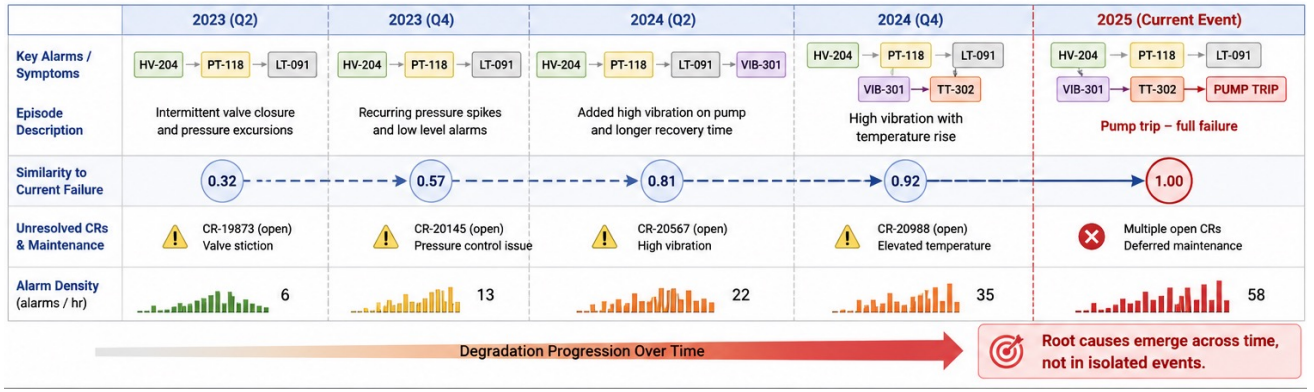


Analytics Methods III: Temporal Pattern Reasoning

- **Alarm fingerprint** (has this happened before?): Compare alarm and event sequences against historical events to identify recurring failure patterns



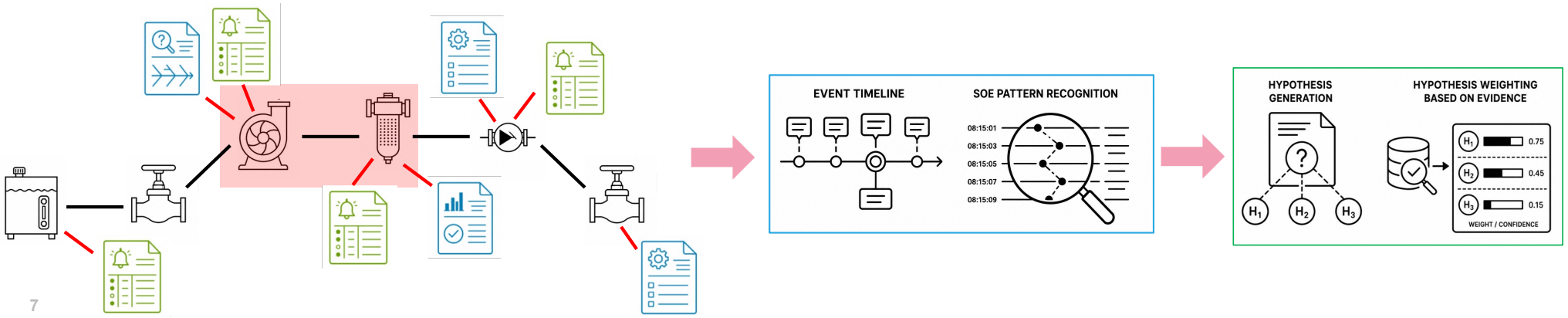
- **Degradation chain** (how did we get here?): Connect related events across time to reveal long-term degradation pathways and recurring issues



Analytics Methods IV: Scoring and Synthesis

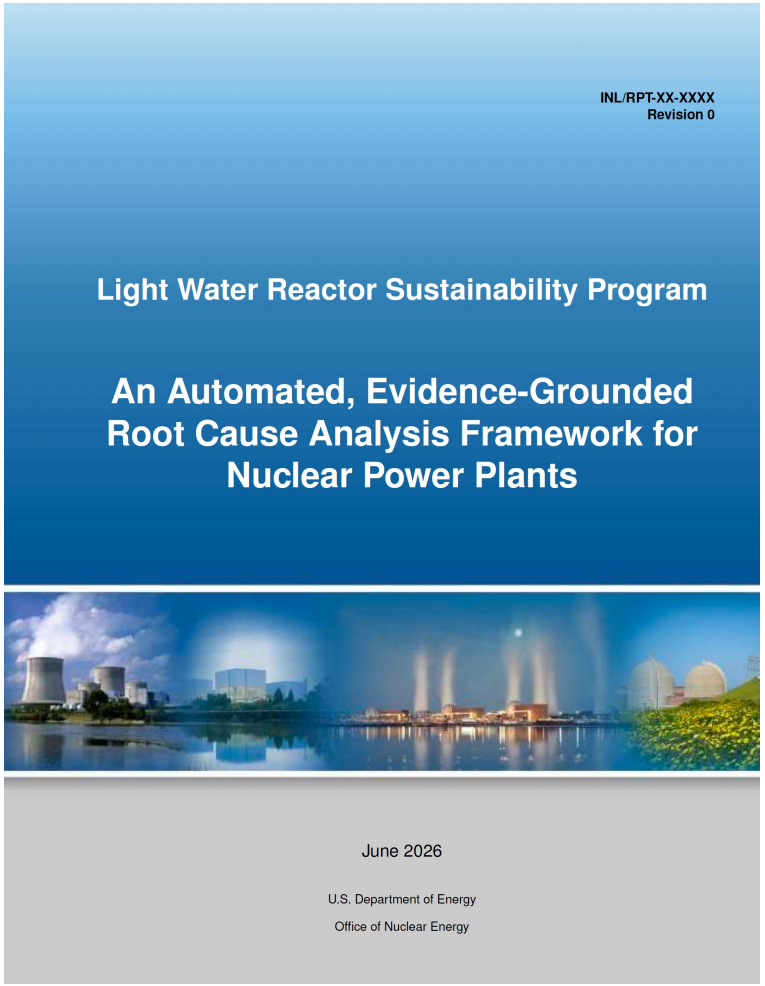
- **Evidence produced by previous analytics engines**
  - Knowledge extraction (what happened?)
  - Temporal analysis (when did it happen?)
  - Pattern recognition (has it happened before?)
- **RCA investigation**
  - Identify hypothesis (what are the suspects?)
  - Evidence integration (what is the evidence basis?)
  - Scoring & synthesis (what does the evidence support?): Integrate multiple lines of evidence to rank the most plausible causes

Three levels of cause
1. Proximate causes (component failure)
2. Contributing causes (maintenance, design, human)
3. Root causes (programmatic and organizational)



FY26 Remaining Work

1. Finalize technical report
2. Complete verification and regression testing
3. Perform initial evaluation using OPG plant data
4. Prepare FY27 utility demonstration
5. Publications
   - PHM conference
   - PSAM conference



FY27 Plan

- Build on top of the knowledge platform implemented in FY26 (RCA reasoning workflow, data analytics engines, and verification cases)
- **FY27:** Demonstrate a plant knowledge management and discovery platform that equips nuclear operators and safety analysts with evidence-backed causal insights to support faster, more defensible decisions
  - Demonstrate on utility data (plant records, historical cases)
  - Independent engineer review
    - Compare platform results with historical investigations
    - Capture analyst feedback
  - Development: Provide analysts with an intuitive interface to explore evidence, evaluate hypotheses, iterate on the reasoning process, and refine RCA conclusions
- **Tools:** DACKAR: https://github.com/idaholab/DACKAR
- **Collaboration:** Ontario Power Generation (share data, provide feedbacks); initial discussion with Xcel and TVA
- **Impact** (ref. EO 14302 goals, UPRISE)
  - Plant reliability: Improves consistency of RCA investigations and reduce risk of unplanned outages
  - Workforce sustainability: Captures and reuses expert knowledge
  - Knowledge management: Turns decades of plant records into usable organizational knowledge

Vision: Capability Expansion Roadmap

- A **plant knowledge management and discovery platform** that integrates plant data, engineering knowledge, and analysis models to provide evidence-based insights for engineering, operational, risk-informed, and regulatory decisions

FY27 Knowledge + Data	**Evidence-based reasoning**	**Validation**: Find evidence-backed causal explanations, support corrective action programs, reduce investigation burden
FY28 Knowledge + Data + Trends	**Knowledge-based reasoning**	**Operational intelligence:** Discover emerging trends, recurring patterns, and hidden relationships across plant knowledge and operating experience
FY29 Knowledge + Data + Models	**Model-based reasoning**	**Configuration assurance:** Discover inconsistencies between plant knowledge, operational data, and engineering models